BioLoop Water Efficiency Model (BWEM)

Prototype Documentation – Version 1.0

1. Overview

The BioLoop Water Efficiency Model (BWEM) Prototype v1.0 is a publicly deployed, planning-level digital decision-support tool designed to estimate farm-level water demand and irrigation efficiency in semi-arid agricultural regions.

The prototype integrates crop evapotranspiration modeling, livestock water demand estimation, and irrigation system efficiency factors into a structured assessment framework accessible through a web interface.

Deployment URL: bioloop.net/water-calculator

Release Version: 1.0

2. Core Functional Components

- The prototype allows users to:
- Input farm area (acres)
- Enter reference evapotranspiration (ETo)
- Select crop type (crop coefficient-based)
- Select irrigation method (flood, sprinkler, drip)
- Enter livestock type and population size
- The system then calculates:
- Crop daily water requirement (ETc = ETo × crop coefficient)
- Livestock daily water demand
- Total farm water demand
- Estimated irrigation efficiency
- Estimated irrigation inefficiency (water loss)
- Basic water management recommendations

3. Methodological Basis

The prototype uses standard agricultural water modeling principles, including crop evapotranspiration estimation using the formula $ETc = ETo \times Kc$.

Where ETo represents reference evapotranspiration, and Kc represents the crop coefficient.

Livestock demand is estimated using species-based daily intake assumptions. Irrigation efficiency adjustments are based on generalized system performance benchmarks.

The model is designed for planning-level analysis rather than regulatory or engineering certification.

4. Calculations and Unit Conversions (BWEM Core Logic)

BWEM computes crop water demand using the FAO-56 crop coefficient method and converts water depth estimates into volumetric irrigation requirements based on irrigated area and irrigation system efficiency. Calculations are performed as follows:

1) Crop evapotranspiration (ETc)

BWEM calculates crop evapotranspiration using:

$ETc\ (mm/day) = ETo\ (mm/day) \times Kc$

where ETo is reference evapotranspiration and Kc is the crop coefficient.

2) Convert irrigated area to square meters

BWEM converts irrigated area from acres to square meters using:

$Area\ (m^2) = Area\ (acres) \times 4{,}046.86$

3) Convert ETc depth to net crop water volume (liters/day)

Because 1 mm of water over 1 m² equals 1 liter, BWEM converts ETc to daily net water volume as:

$Net\ crop\ demand\ (L/day) = ETc\ (mm/day) \times Area\ (m^2)$

4) Convert liters to gallons (gallons/day) BWEM converts liters to U.S. gallons using:

Net crop demand (gal/day) = Net crop demand (L/day) ÷ 3.78541

5) Adjust for irrigation efficiency to estimate applied irrigation requirement

BWEM estimates the applied irrigation demand by adjusting for irrigation system efficiency (Ei, expressed as a decimal).

Applied irrigation (gal/day) = Net crop demand (gal/day) ÷ Ei

Examples used in pilot scenarios include: flood (Ei = 0.60), sprinkler (Ei = 0.75), and drip (Ei = 0.90).

6) Estimate irrigation loss (gal/day)

BWEM estimates loss as the difference between applied irrigation and net crop demand:

Irrigation loss (gal/day) = Applied irrigation (gal/day) − Net crop demand (gal/day)

7) **Livestock water demand**

When livestock demand is included, BWEM calculates:

Livestock demand (gal/day) = Intake rate (gal/day/head) × Headcount

8) Total estimated farm water demand (gal/day)

BWEM reports total daily farm demand as:

Total demand (gal/day) = Applied irrigation (gal/day) + Livestock demand (gal/day)

(If no livestock is included, total demand equals applied irrigation.)

Constants used:

- 1 acre = 4,046.86 m²
- 1 mm over 1 m² = 1 liter
- 1 gallon = 3.78541 liters

5. Demonstration of Operational Functionality

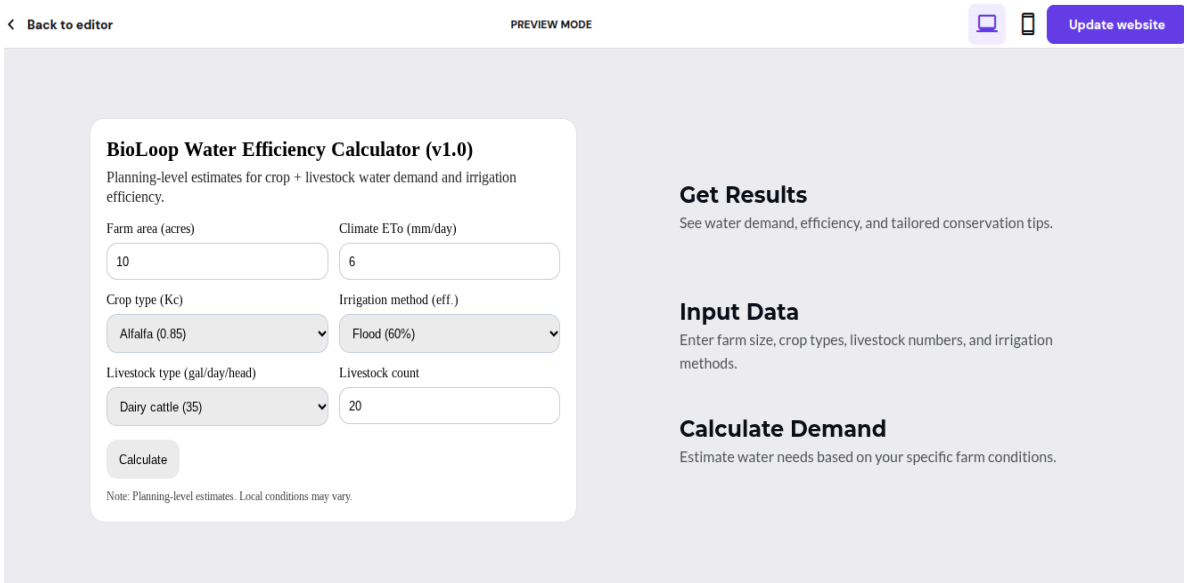
The deployed system performs real-time numerical calculations based on user inputs, adjusts outputs dynamically when irrigation efficiency changes, generates efficiency percentages and estimated water losses, and provides conditional recommendations based on climate and irrigation type.

The system has been tested on desktop and mobile interfaces to ensure usability and responsiveness.

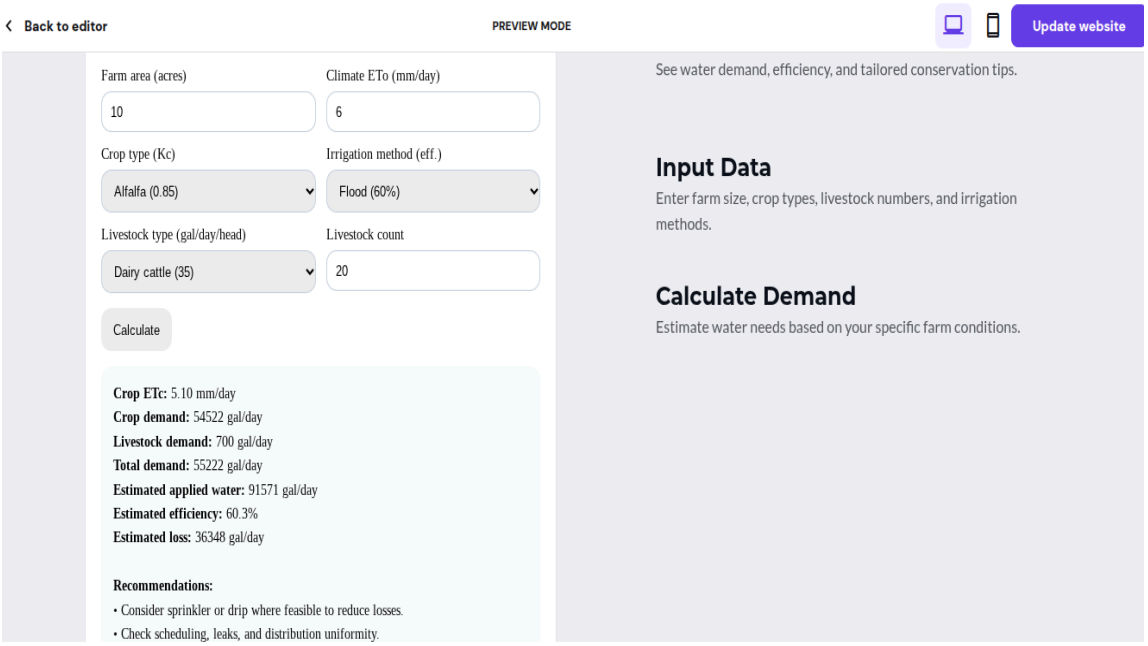
Pilot simulations demonstrating operational outputs are provided in **(Exhibits E1-E2)**

Screenshots of input interface and output results (Prototype v1.0):

- Screenshot of the input interface.



- Screenshot of output results.



6. Development Scope and Future Enhancement

Version 1.0 represents the foundational phase of the BioLoop framework.

- Planned enhancements may include:
- Integration of real-time weather APIs
- Incorporation of soil classification parameters
- Enhanced livestock temperature modeling
- Scenario comparison modules for irrigation transition analysis
- Expansion into broader circular agricultural optimization systems

The prototype demonstrates technical feasibility and establishes the operational foundation for scalable agricultural water efficiency modeling.

BWEM:

Link: https://bioloop.net/water-calculator

Current Functionality and Next Validation Steps

The **BioLoop Water Efficiency Model (BWEM)** is currently developed as a planning-level decision-support tool designed to estimate agricultural water demand and support irrigation-related decision-making. The model integrates crop water requirements based on evapotranspiration principles, irrigation efficiency parameters, and simplified farm-level inputs to generate comparative outputs under different irrigation scenarios.

At its current stage, BWEM is capable of:

- Estimating crop water demand using established methodologies (e.g., FAO-56 approach)
- Comparing irrigation methods (e.g., flood, sprinkler, drip)
- Estimating water losses associated with different irrigation efficiencies
- Providing structured outputs to support planning decisions

The model has been implemented in a functional prototype and is accessible through an online interface, demonstrating its operational feasibility as a digital tool.
At its current stage, The prototype is intended to support pre-season planning and comparative evaluation of irrigation strategies in water-limited agricultural environments.

Next validation steps include:

- Collecting feedback from subject-matter experts in agricultural water management
- Refining model assumptions and improving clarity of outputs
- Evaluating usability and applicability in real-world planning contexts
- Exploring opportunities for pilot-level use or advisory application

These steps are intended to further strengthen the model's reliability and practical relevance.

Use Limitations

BWEM is designed as a planning-level analytical tool and should be interpreted within the scope of its intended use. The model is not intended to replace detailed engineering analysis or real-time irrigation management systems.

Key limitations include:

- The model operates on simplified assumptions and generalized input parameters
- It does not incorporate real-time environmental data or dynamic field variability
- It does not account for detailed soil heterogeneity or microclimate variations
- It is not an engineering-certified system for operational irrigation control
- The outputs are intended for comparative planning and estimation purposes only

These limitations are acknowledged to ensure accurate interpretation of results and to maintain transparency regarding the model's scope.

Despite these constraints, BWEM provides meaningful insights at the planning stage, particularly in evaluating relative differences between irrigation strategies and identifying opportunities for improved water-use efficiency.

Planned Stakeholder Validation

To enhance the robustness and applicability of BWEM, future development will include structured engagement with relevant stakeholders in the agricultural and environmental sectors.

Planned stakeholder validation activities include:

- Engaging agricultural advisors, water management professionals, and extension specialists
- Sharing the model for feedback on usability, assumptions, and practical relevance
- Collecting qualitative input on real-world applicability and decision-making value
- Identifying potential improvements based on user experience and expert recommendations

This validation process is intended to ensure that the model aligns with real-world needs and reflects practical constraints faced by agricultural stakeholders.

Initial outreach efforts, including communication with subject-matter experts and feedback requests, have already been initiated.

Through this approach, BWEM will continue to evolve as a user-informed decision-support tool, with the goal of improving water-use efficiency and supporting sustainable agricultural practices, particularly in water-limited regions.